

Observing LLM API Latency During Peak and Off-Peak Hours: A 28-Day Log

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Abstract

Commercial large language model (LLM) APIs are shared infrastructure, yet little is known about how server load during peak business hours affects latency, throughput, and answer quality in practice. We benchmark four providers—OpenAI (`gpt-4o-mini`), Anthropic (`claude-haiku-4-5`), Google (`gemini-flash-lite`), and Alibaba Qwen (`qwen3.5-flash`)—twice daily for 28 days using 30 MMLU-Pro questions, measuring time-to-first-token (TTFT), tokens-per-second (TPS), and answer accuracy. Peak sessions run at 10:00 ET (within the 8 AM–2 PM ET peak band [Anthropic, 2026]); non-peak sessions run at 22:00 ET. Analyzing at the level of complete measurement slots—the appropriate unit of statistical independence—we find that peak-hour TTFT is directionally higher for the three US-hosted providers (+3.5% to +18.1% in slot medians), though this effect is only marginally significant and, for Google, not robust to a magnitude-based test. The most robust result is a reversed pattern for Qwen—served from an Alibaba Cloud US endpoint—which is *faster* during US peak hours (−9.4% TTFT, +13.4% TPS; significant under both rank- and magnitude-based tests), consistent with the hypothesis that US peak corresponds to Asia off-peak. Answer accuracy shows no significant change for any provider across conditions, suggesting that output quality is decoupled from infrastructure load at the scale tested. We emphasize that with only nine complete slots the study is underpowered; we therefore report effect sizes as a directional pilot signal rather than confirmatory findings.

1 Introduction

Large language model APIs have become commodity infrastructure for production applications,¹ yet their service-level properties under varying load conditions remain largely undocumented from the consumer’s perspective. Traffic on commercial AI APIs follows predictable diurnal patterns driven by North American business hours, and the substantial per-request compute of LLM inference may amplify the effect of concurrent demand. If providers do not over-provision sufficiently, practitioners who call APIs during peak hours may receive degraded latency—increasing costs for time-sensitive applications such as real-time agents or interactive chat. Whether throughput degradation also affects output *quality* (answer accuracy) is an open question with direct implications for reliability-critical deployments.

This paper reports the results of a 28-day empirical study comparing peak-hour and off-peak-hour performance across four major LLM API providers on a standardized MMLU-Pro benchmark suite. Our primary contributions are:

1. A reproducible measurement framework for longitudinal LLM API benchmarking.

¹The English writing in this paper was assisted by Claude (Anthropic).

2. Provider-specific quantification of peak-vs-nonpeak latency and throughput effects.
3. Evidence that MMLU-Pro accuracy is robust to infrastructure load conditions.
4. An anomalous finding for a non-US-headquartered provider suggesting inter-regional load balancing.

We regard this work as a pilot study. Limitations in complete-slot coverage, question-set size, and network control are acknowledged; a more rigorous follow-up study is planned.

2 Methodology

2.1 Providers and Models

We evaluated four providers and their smallest fast-inference models available at the time of the study (Table 1). We specifically chose lower-cost, high-throughput models under the hypothesis that budget-tier endpoints—which are more likely to be densely shared—would exhibit greater sensitivity to peak-hour load than premium offerings. All models were prompted with a fixed system prompt instructing single-letter responses (A–D), with `temperature=0` for reproducibility. Streaming inference was used for all calls to measure TTFT directly.

Table 1: Models under evaluation.

Provider	Model	Endpoint region
OpenAI	<code>gpt-4o-mini-2024-07-18</code>	US East
Anthropic	<code>claude-haiku-4-5-20251001</code>	US East
Google	<code>gemini-3.1-flash-lite-preview</code>	US (GCP)
Qwen	<code>qwen3.5-flash</code>	US (Alibaba Cloud)

2.2 Question Set

We drew 30 questions from the *hard* split of MMLU-Pro [Wang et al., 2024], an extension of the original MMLU [Hendrycks et al., 2021] that increases difficulty and expands each item from four to up to ten answer options. Our sample is balanced across six subjects (mathematics, physics, engineering, history, law, and psychology), five questions each; 27 of the 30 items have ten options and three have nine. Each question was repeated three times per session to reduce variance, yielding 90 trials per provider per session (30 questions $\times$ 3 repeats). Accuracy was scored by extracting the first valid option letter from the model response using a cascade of regular expressions targeting XML-tagged answers, explicit “Answer:” labels, and trailing isolated letters. We verified that this parser is not a bottleneck: format failures (responses with no parseable option letter) are rare (Section 3.4), and a lenient variant that additionally maps an answer’s option *text* back to a letter recovers essentially no additional correct answers. Reported accuracy therefore reflects genuine option selection, not parsing loss.

2.3 System Prompt

All providers received the same system prompt, which instructed the model to (1) write step-by-step reasoning inside a `<Reason>` tag and (2) output only a single correct option letter inside an `<Answer>` tag. No additional context or few-shot examples were provided. The prompt was held constant across all sessions and providers to ensure comparability.

2.4 Experimental Schedule and Session Flow

Experiments ran from 2026-03-30 to 2026-04-26 (28 days). Two sessions were scheduled daily in the America/New_York timezone:

- **Peak:** 10:00 ET — within the 8 AM–2 PM ET peak band [[Anthropic, 2026](#)].
- **Non-peak:** 22:00 ET — US East Coast late night.

Sessions were managed by a Python scheduler (APScheduler) running as a launchd daemon on a Mac Mini with a residential internet connection.

Within each session, questions were iterated in a fixed order. For each question, all four providers were called sequentially in a fixed order (OpenAI → Anthropic → Google → Qwen), with a 2-second pause between calls to avoid triggering rate limits. This cycle was repeated three times per question (REPEATS=3).

2.5 Resumption Logic

Each individual API call was recorded as a task in a JSON progress file upon completion. If a session was interrupted (e.g., due to an API error or system restart), the scheduler detected the unfinished session on the next run and resumed from the last completed task, skipping already-recorded calls. Sessions marked as fully complete were not re-executed.

2.6 Metrics

- **TTFT** (time-to-first-token, seconds): wall-clock time from request dispatch to receipt of the first streamed token.
- **TPS** (tokens per second): output token count divided by generation time after the first token.
- **Accuracy:** proportion of trials where the parsed answer matches the correct MMLU-Pro key.

2.7 Statistical Analysis

The unit of statistical independence is the measurement *slot*, not the individual API call: the 360 calls within a slot share that session’s server-load and network state and are therefore strongly correlated. Testing at the call level would treat these correlated observations as independent (pseudoreplication), inflating the effective sample size by more than two orders of magnitude and yielding spuriously small p -values. We therefore first aggregate each metric to a single value per (slot, provider)—the median for latency and throughput, the mean for accuracy and format-failure rate—and then compare peak slots with non-peak slots. Because only a few slots per condition survive the completeness filter (Section 5), we report two complementary tests: a two-sided exact Mann–Whitney U test on slot values (p_{exact} , rank-based) and an exact permutation test on the difference in slot means (p_{perm} , magnitude-based and more sensitive to outlier slots). Effect sizes are reported as slot-level differences and percent change relative to the non-peak value. We treat $p < 0.05$ as the significance threshold but, given the slot count, interpret results as suggestive rather than confirmatory. A linear mixed-effects model with a per-slot random intercept—the principled way to retain call-level data—could not be reliably fit at the current slot count and is deferred to a larger follow-up.

3 Results

3.1 Data Completeness

Of 56 scheduled slots (28 days $\times$ 2 periods), 54 were executed; 3 non-peak slots were missed due to daemon downtime. Most executed slots were partial: API errors (primarily Google returning HTTP 429/503) caused premature termination in 45 of 54 slots. A *complete* slot is defined as exactly 360 rows (30 questions $\times$ 3 repeats $\times$ 4 providers). Only 9 slots were complete (3 non-peak, 6 peak). All primary results below use these 9 complete slots (3,240 rows total, aggregated to $9 \times 4 = 36$ slot-level observations); see Section 5 for discussion of this constraint.

3.2 Latency: Time-to-First-Token

Table 2: TTFT: peak vs. non-peak at the slot level (complete slots only; $n_{\text{peak}} = 6$ slots, $n_{\text{nonpeak}} = 3$ slots; slot medians reported). p_{exact} : two-sided exact Mann–Whitney on slot medians. p_{perm} : exact permutation test on the mean slot difference.

Provider	Non-peak (s)	Peak (s)	Δ (s)	Δ (%)	p_{exact}	p_{perm}
OpenAI	0.876	0.906	+0.031	+3.5%	0.167	0.417
Anthropic	0.828	0.923	+0.095	+11.5%	0.024	0.060
Google	1.689	1.996	+0.306	+18.1%	0.048	0.345
Qwen	3.722	3.372	−0.350	−9.4%	0.048	0.048

All three US-hosted providers show directionally higher TTFT during peak hours (Table 2), from +3.5% (OpenAI) to +18.1% (Google) in slot medians. The statistical evidence, however, is weak once the slot is treated as the unit of independence: only Anthropic and Google reach $p < 0.05$ under the rank-based exact test, and Google’s effect does not survive the magnitude-based permutation test ($p_{\text{perm}} = 0.35$), indicating it is driven by a few high-latency slots rather than a consistent shift. OpenAI’s increase is not significant under either test. Qwen is the clear exception: its TTFT is *lower* during peak hours, and this reversal is significant under both tests—the most robust latency effect in our data—a pattern we discuss in Section 4.

3.3 Throughput: Tokens per Second

Table 3: TPS: peak vs. non-peak at the slot level (complete slots only; slot medians; $n_{\text{peak}} = 6$, $n_{\text{nonpeak}} = 3$ slots). Columns as in Table 2.

Provider	Non-peak	Peak	Δ	Δ (%)	p_{exact}	p_{perm}
OpenAI	55.1	51.8	−3.3	−5.9%	0.381	0.286
Anthropic	141.2	138.2	−3.1	−2.2%	0.548	0.452
Google	225.7	229.8	+4.0	+1.8%	0.548	0.631
Qwen	167.9	190.5	+22.5	+13.4%	0.048	0.024

TPS differences are small and non-significant for the three US-hosted providers under both tests (Table 3). Qwen again shows the opposite pattern: generation throughput is significantly higher during US peak hours (+13.4%, significant under both the rank- and magnitude-based tests).

Table 4: MMLU-Pro accuracy: peak vs. non-peak at the slot level (mean correct per slot, complete slots only; $n_{\text{peak}} = 6$, $n_{\text{nonpeak}} = 3$ slots).

Provider	Non-peak	Peak	Δ	p_{exact}
OpenAI	0.493	0.480	-0.013	0.548
Anthropic	0.833	0.817	-0.017	0.262
Google	0.867	0.865	-0.002	0.905
Qwen	0.822	0.811	-0.011	0.548

3.4 Answer Accuracy

No provider shows a statistically significant change in accuracy between peak and non-peak conditions (Table 4, all $p > 0.25$). Absolute accuracy varies widely across providers, from $\approx 48\%$ for OpenAI’s `gpt-4o-mini` to $\approx 87\%$ for Google’s `gemini-flash-lite`. The low `gpt-4o-mini` figure is genuine performance on the hard, ten-option MMLU-Pro split rather than a measurement artifact: format-failure rates are low (2.3% for OpenAI, below 1% for the others), accuracy computed only over parse-successful responses is essentially identical to the overall figure (0.496 vs. 0.484 for OpenAI), and even an upper bound that credited *every* unparseable response as correct would raise OpenAI’s accuracy only to $\approx 51\%$. The result is consistent with a small, budget-tier model on a deliberately difficult benchmark.

4 Discussion

US-hosted providers: TTFT trends upward with load. OpenAI, Anthropic, and Google all show elevated slot-median TTFT during peak hours, directionally consistent with increased queuing or scheduling overhead under higher concurrent demand. The point estimates range from $\approx 3.5\%$ for OpenAI to $\approx 18\%$ for Google, which could reflect different capacity margins across providers. We stress, however, that with three to six slots per condition this trend is only weakly supported: OpenAI’s increase is not significant, and Google’s—though the largest—does not survive a magnitude-based test, indicating sensitivity to a small number of high-latency slots. These effects should be read as hypotheses to be confirmed with the expanded slot coverage outlined in Section 5. Notably, whatever latency increase exists does not propagate to generation speed or accuracy, implying that once the model begins generating, execution is largely isolated from request-level congestion.

Qwen anomaly: possible effect of regional user distribution. Qwen’s US endpoint (`dashscope-us.aliyuncs.com`) is operated by Alibaba Cloud, whose primary user base is concentrated in Asia. US business peak hours (10:00 ET) correspond to approximately 23:00–01:00 CST the following day, well outside Asian business hours. The fact that Qwen showed lower TTFT and higher TPS precisely during US peak hours is consistent with the interpretation that US-region capacity is relatively underutilized when Asian demand is low—though we cannot confirm this from API-level measurements alone.

Accuracy robustness. The absence of accuracy degradation during peak hours is practically important. Applications relying on LLM APIs for classification or reasoning can expect consistent output quality regardless of the time of day. Operators optimizing purely for quality need not schedule around traffic patterns. Those optimizing for latency-sensitive workloads (real-time agents, interactive chat) may see TTFT increases on the order of 3–18% during US peak hours for US-hosted providers, though our data can only establish this trend directionally.

5 Limitations

Limited complete-slot coverage and Google data gaps. Of 54 executed slots, only 9 met the completeness threshold of 360 rows. The primary cause was Google Gemini API rate-limiting (HTTP 429/503 errors), which terminated experiments mid-session, resulting in entire Google response rows being absent rather than malformed. This disproportionately affected non-peak slots (only 3 complete) and may introduce selection bias. Analyses are restricted to complete slots to minimize this bias, but the sample covers only a subset of the 28-day window and severely limits statistical power.

Low statistical power and the slot as the unit of analysis. Because calls within a slot are correlated, the effective sample size is the number of complete slots (nine), not the number of calls. With three to six slots per condition, the smallest attainable two-sided exact p -value is ≈ 0.024 , so even our “significant” effects sit near this floor and cannot be treated as confirmatory. We report two tests (rank- and magnitude-based) precisely because they can disagree at this scale, and we interpret all latency and throughput effects as directional pilot signals. Confirmatory analysis—including the per-slot random-intercept mixed model we could not fit here—requires substantially more complete slots.

Residential network variability. All measurements were made from a Mac Mini on a residential internet connection. Network jitter contributes to TTFT variance and cannot be fully separated from server-side latency effects.

Small question set and statistical power. Each question was repeated three times per session to reduce per-question noise, but the pool of 30 MMLU-Pro questions remains small. This limits coverage across subjects and is insufficient to detect small accuracy effects ($< 2\%$) or subject-level differences. A larger and more diverse question set would improve sensitivity.

Non-comparable generation speed across providers. Qwen (qwen3.5-flash) returns extended chain-of-thought reasoning by default, producing responses substantially longer than those of other providers for the same prompt. As a result, TPS figures for Qwen reflect a different output regime and direct cross-provider speed comparisons are not meaningful. Within-provider peak vs. non-peak TPS comparisons remain valid.

Model versioning. Providers may silently update model weights or infrastructure during the study period. Observed trends reflect the combined effect of load patterns and any within-study infrastructure changes, which we did not control for.

6 Conclusion

We conducted a 28-day longitudinal benchmark of four commercial LLM APIs at US business-hour peak and off-peak times. Analyzed at the slot level, US-hosted providers (OpenAI, Anthropic, Google) show directionally elevated TTFT during peak hours (3–18% in slot medians), with no corresponding degradation in generation throughput or answer accuracy; this trend is only marginally significant given nine complete slots. The most robust finding is that Qwen, served from a US-region Alibaba Cloud endpoint, shows the opposite pattern—faster TTFT and higher TPS during US peak—significant under both tests and consistent with cross-regional load balancing.

These findings suggest that practitioners may expect modest latency increases for latency-critical applications during US business hours but can rely on consistent output quality regardless of time of day, with the important caveat that the latency effects are directional pilot signals

rather than confirmed results. This study should be regarded as a pilot investigation. The limited number of complete slots, small question set, and uncontrolled network conditions constrain the generalisability of the results. A follow-up study will address these shortcomings by mitigating API rate limits, controlling for network variability, expanding the question pool, and adopting more rigorous evaluation metrics to place the findings on a firmer empirical footing.

The English writing in this paper was assisted by Claude (Anthropic).

Data and Code Availability

The benchmark framework, collected CSVs, and analysis scripts are available at https://github.com/silverbreak1/llm_traffic_router.

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